

Nathan Tang

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Canadian citizen eligible for U.S. employment with TN status, open to relocation.

EXPERIENCE

People Inc. (Dotdash Meredith) *Software Engineer*

May 2022 – Present

- One of three founding engineers on a full-stack ad configuration platform (Node.js, Java, PostgreSQL, Vue.js), enabling instant global updates across the ad network, contributing **\$1M/day in revenue**.
- Led backend architecture and service design, scaling Fastify APIs to handle **millions of requests daily**, including schema design, query optimization, and authentication systems.
- Built reusable components adopted **across 40+ brands**, enabling flexible configuration and improving viewability while **sustaining 95+ on Core Web Vitals** scores.
- Engineered the backend data processing pipeline for dynamic bundling, recursive configuration sorting, and configuration tiering, **reducing payload size by 30%** and improving API response times.
- Implemented a deduplication and tiering system that merged overlapping ad configurations, **reducing database footprint by 40%** and server load substantially.
- Built tooling to automate API schema-to-DTO generation, creating type-safe backend–frontend interfaces, accelerating development, and reducing runtime errors through consistent contracts.
- Developed OAuth-secured configuration CMS, streamlining ad operations workflows for internal teams.

Aquatic Biosphere Society of Canada *Full Stack Engineer*

January 2024 – April 2024

- Designed and implemented APIs (Python, PostgreSQL) for the full gameplay loop, scoring, user tracking, and CMS functionality, integrating both the game and admin tools with Google OAuth authentication.
- Developed OpenCV based backend pipeline, **achieving <1 second image processing (down from 10 seconds)** through optimization for real-world bandwidth and device constraints.
- Built a custom admin CMS interface (Node.js, Svelte) for content teams to manage game types, game content, facts, scores, and user data with real-time updates and live content editing.
- Designed normalized database schemas for user profiles, game sessions, and mural metadata, supporting future extensibility for new game types and media assets.
- Deployed on Azure, using web and database services to support live gameplay and image workloads.

Snap & Tow *Backend Engineering Intern*

March 2022 – May 2022

- Built secure authentication flows using JWT and SendGrid for sign-up, login, and password recovery.
- Integrated with Google Maps API to match users with the closest operators in real time based on location.

PROJECTS

thechoice.fm | PHP, JavaScript, Node.js, MySQL

- Developing a real-time on-demand web radio application enabling synchronized playback for all users via a distributed playlist system, with a shared queue experience.
- Building an API + dashboard to manage users/queues, interfacing directly with radio automation software.

Discord Bot | TypeScript, Node.js, Docker, MySQL

- Developed a bot that queries game servers in real time using the UDP Query protocol to return live status and player information. Designed for resiliency using Docker for isolation and persistent state.
- **Deployed to 700+** Discord communities and serves **hundreds of active users** daily.

EDUCATION

University of Alberta | Bachelor of Science in Computing Science

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, Go (Golang), Java, C, HTML/CSS, SQL
Databases	MySQL, PostgreSQL, MongoDB, Redis, Firebase
Technologies	Node.js, React, Vue, Svelte, Tailwind, Express.js, Django, Git, Docker, AWS, Azure